GUARDIAN

MAKERS



Installation and user manual

Technical documentation
English edition, October 2026

Operation, wiring and maintenance
Italian / English interface

Reading guide

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This guide describes the bilingual GUARDIAN interface. The owner reports successful initial operation. That functional result is distinct from measurement validation.

Original display photographs retain their Italian text. English headings and instructions explain the corresponding functions. The cover image is retained from the approved presentation.

Purpose and limits

For maker workspaces

GUARDIAN tracks the environment and displays information from one 3D printer. It helps compare periods and retain readings with their printing context.

Scope

It is an environmental monitor. It is not a reference instrument, a certified safety detector or a replacement for ventilation and exposure assessment.

Printer compatibility

Testing used a Bambu Lab A1. The integration follows Bambu Lab protocols. Other Bambu Lab models require verification; other brands require investigation and possible adaptation.

Evidence in this guide

The photographs show actual prototype readings over partial test periods, not complete 24-hour averages. Network details and credentials are examples. English browser views are labelled separately.

Quick start

1. Position

Place GUARDIAN on a stable surface with the sensor air openings unobstructed. Keep it away from local heat, breathing and direct extraction jets.

2. Power

Use the rear USB-C power input in normal operation. Keep the side programming port covered by the TPU cap. Do not connect the two power sources at the same time.

3. Check measurements

Allow sensor startup and stabilisation. Check HOME and the sensor status. Missing readings do not mean zero.

4. Configure

Set Wi-Fi, time zone, personal thresholds and display lighting. Set up the printer with its local IP address, serial number and LAN access code.

5. Record

Insert and test a compatible microSD card with power off. Confirm that recording rows increase, then download the current CSV from the browser.

Hardware and assembly

Main components

Freenove ESP32-S3 5-inch display, Sensirion SEN69C, MP1585EN converter, USB-C adapter, rear power socket and microSD card. MP1585EN is the module designation confirmed by the builder.

Enclosure

Keep the sensor low at the rear, on its removable support. The internal partition separates it from the electronics. Keep inlet and outlet paths distinct and unobstructed.

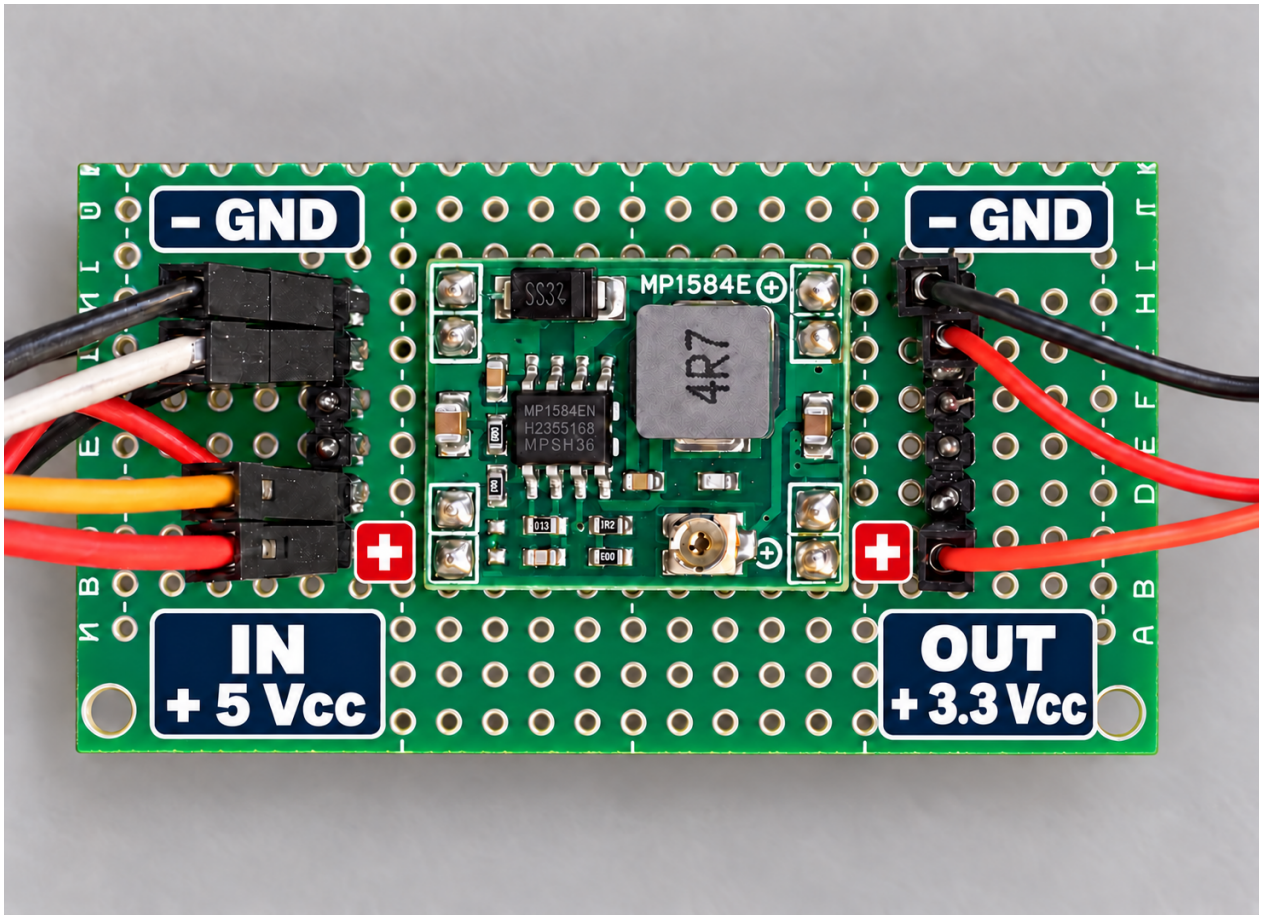
Fasteners

Display and USB board: M2.5 × 5 mm button-head screws. Internal partition and sensor panel: M2.5 × 6 mm countersunk Phillips screws. Case/front panel: M3 × 30 mm socket-head screws.

Inserts and supports

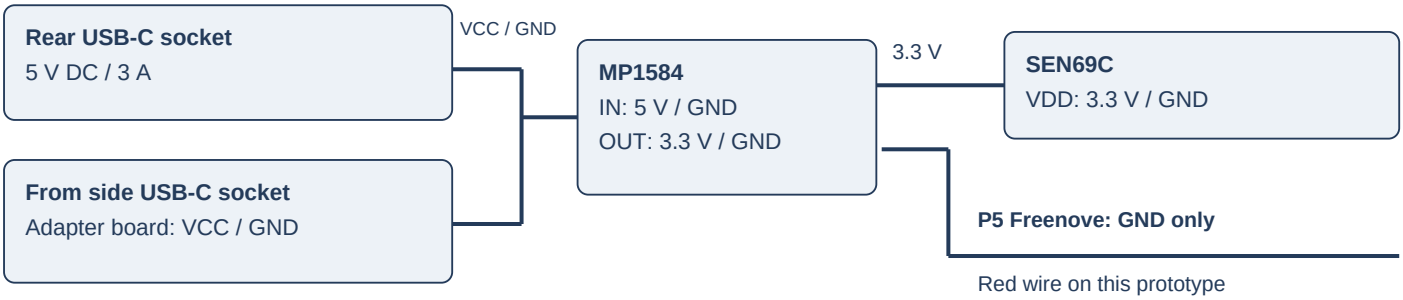
Use the supports and insert seats in the final mechanical files. Do not force screws or overtighten inserts in printed plastic. Verify the assembled clearances before powering the device.

Power supply and wiring



Perfboard 60 x 44 mm | MP1584 22.3 x 16.8 mm | Male Dupont headers

Edited photograph: illustrative labels. Identify terminals from actual board markings.



One power source at a time. Rear and side sockets are directly paralleled, without isolation. Disconnect rear power before connecting the PC to the side socket; disconnect the PC before restoring rear power. The TPU cap identifies the programming port; it is not an electrical interlock.

Perfboard connections. Rear VCC and VCC from the side adapter board connect to IN+; both grounds connect to IN-. OUT+ at 3.3 V supplies SEN69C VDD; OUT- connects to sensor GND and Freenove common ground. Measure and adjust the output before connecting the sensor. Never supply it directly with 5 V.

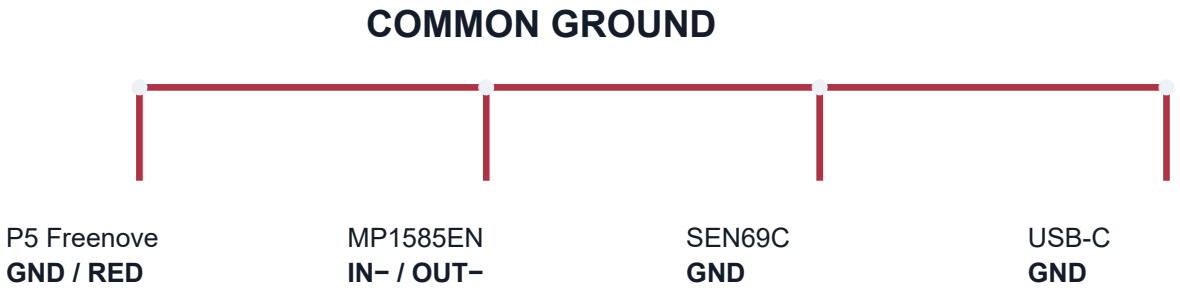
SEN69C signals [S1]. Pin 1: VDD; pin 2: GND; pin 3: SDA to GPIO19 (P4); pin 4: SCL to GPIO20 (P4). Pins 5 and 6 duplicate GND and VDD: follow the manufacturer connector drawing. P5: use GND only; insulate the other three wires. Do not bridge D+ and D-.

Power section update: MP1584, as specified by the builder on 22 October 2026.

P5: common ground only

RED WIRE = GND (NEGATIVE)

Use only the RED GND wire from P5. Leave the other three P5 wires disconnected and individually insulated. The sensor receives power from MP1585EN, not from P5.



Other P5 pins: NOT CONNECTED

P5 3.3 V, IO17 and IO18 are unused in this connection. Follow the board labels, not wire colours. The red/GND identification applies to the cable photographed and confirmed by the builder.

Sensor power

MP1585EN OUT+ adjusted to 3.3 V connects to SEN69C VDD. OUT− connects to sensor GND and the common ground. Check output voltage before connecting the sensor.

Functional ground diagram only. Use one power source at a time; rear and side USB power inputs are directly paralleled.

Adapter and converter installation

P5 ground

Only the RED wire at P5 GND is used for the common ground. P5 3.3 V, IO17 and IO18 remain disconnected and individually insulated. The colour identification applies to the confirmed prototype cable.

Module pin spacing

The MP1585EN module used here has a slightly wider pin pitch than the perfboard grid. Check alignment, use short connections and do not force its pins into misaligned holes.

USB data

Do not bridge D+ and D−. The added jumpers were removed from the prototype adapter to restore communication. Power bridges concern only VCC and GND.

Pull-ups

The Freenove configuration used in this project already includes I2C pull-ups to 3.3 V. Check the actual board before adding parallel resistors. Do not connect I2C pull-ups to 5 V.

One power source

The TPU cap identifies and covers the programming port. It is not an electrical interlock. Remove rear power before attaching a PC, then disconnect the PC before restoring rear power.

Arduino installation

Keep the tested versions

Board package esp32 by Espressif Systems: 3.2.0. Libraries: GFX Library for Arduino 1.6.5; lvgl 9.5.0; TAMC_GT911 1.0.2; Sensirion I2C SEN69C 1.1.0; Sensirion Core 0.7.3; PubSubClient 2.8; ArduinoJson 7.4.3.

Board Manager

Add https://espressif.github.io/arduino-esp32/package_esp32_index.json to Additional Boards Manager URLs, then install esp32 version 3.2.0. [S17]

Library Manager

Select the stated version of each library. Adafruit GFX is not a substitute for GFX Library for Arduino. The IPS display uses GT911 touch. No audio library is required.

Avoid automatic upgrades

Do not use Update All for a working installation. New-version notifications are not faults. Test one change at a time and retain a working copy.

Arduino project and settings

Board options

ESP32S3 Dev Module; OPI PSRAM; Upload Speed 115200; Partition Scheme Huge APP / huge_app. Serial monitor: 115200 baud. Use the remaining Flash/USB options from the working firmware package.

Complete sketch folder

Extract the whole firmware package. Open the .ino file whose name matches the folder. Retain all guardian_* files and bundled resources.

LVGL configuration

Use the working lv_conf.h for LVGL 9.5.0. In the standard arrangement it sits alongside the lvgl library folder. Check the actual include path in compiler output.

Core components

WiFi, NetworkClientSecure, WebServer, ESPmDNS, Preferences, Wire, SPI, FS and SD come with the ESP32 core. FatFs, FreeRTOS and mbedTLS are also included. Do not install competing copies.

Configuration record

Archive the IDE version, every board option and the compile summary. The list here does not replace the complete tested Flash/USB configuration.

Upload and functional checks

1. Back up

Save the existing sketch, lv_conf.h, library versions, settings and CSV files. A normal upload does not format the microSD.

2. Connect

Disconnect rear power, remove the TPU cap and connect a USB data cable to the side port. Select the serial port that appears.

3. Compile

Enable verbose compiler output and use Verify. Check resolved library paths, especially if duplicates exist. Do not upload after a failed compilation.

4. Upload

Close other serial monitors, upload and inspect the startup log at 115200 baud. Confirm the expected firmware version.

5. Check

Test touch, HOME readings, sensor status, SD row growth, CSV download, browser access, lighting and printer data. Retain the previous firmware for rollback.

Updates

Updates currently use USB. OTA updates are not implemented. Preserve flash settings for an ordinary update; RAM history restarts after reboot.

Language selection

Select on the display

Open OTHER → SYSTEM, or ALTRO → SISTEMA while in Italian. Choose English or Italiano. The setting is saved without restarting GUARDIAN.

Browser behaviour

PC and smartphone pages follow the display language and reload within approximately three seconds while reachable. Save open forms first: reloading discards unsaved edits.

Persistence

The selected language survives restart. A factory reset returns to Italian. User-entered names, job names and CSV structure are unchanged. Some serial diagnostics remain Italian.

HOME and navigation



Readings

Each tile shows a value, unit and status. An unavailable reading is not a zero measurement.

Navigation

HOME, CHARTS, PRINTER 3D, ALERTS and OTHER provide the main routes. OTHER opens the sensor, network, maintenance, lighting, diagnostics and system pages.

Comfort

Temperature and humidity comfort bands are personal preferences. Outside comfort is distinct from an environmental alert.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Environmental charts



Parameter and period

Choose a parameter and a 2, 6, 12 or 24-hour view. A light grid helps comparison and the vertical axis uses integer labels.

Details below the chart

Current readings and relevant statistics remain available. Integer axis labels do not remove decimals from measurements or CSV data.

History

Chart history resides in RAM and is interrupted by restart. Existing CSV files remain on microSD.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Measurements and averages

Particulate matter

PM1, PM2.5, PM4 and PM10 are optical estimates of mass concentration in µg/m³. WHO 2021 guideline references for PM2.5 and PM10 are 15 and 45 µg/m³ over 24 hours. PM1 and PM4 use personal thresholds. [S1, S6]

A complete 24-hour average

It requires 24 hours of recording with valid data. Before that, the average covers only the available period and is partial. Check missing samples and interruptions. An internal average does not establish regulatory compliance.

HCHO

Formaldehyde is shown in ppb with a 30-minute average. The WHO 2010 reference is 0.1 mg/m³ over 30 minutes, approximately 81 ppb at 25 °C and 1 atm. The conversion depends on temperature and pressure. [S7]

Other channels

CO2 is shown in ppm as a ventilation indicator. VOC and NOx are relative indices, not individual compound concentrations. Temperature and relative humidity use °C and % RH.

Limits

The monitor does not identify all pollutants or directly measure the full ultrafine particle fraction. Low readings alone do not guarantee safe air. [S1, S8, S9]

Thresholds and comfort



Fast adjustment

Use −100, −10, −1, +1, +10 and +100. Holding a button repeats the change; ±1 accelerates while held. Apply/save and check the result.

Browser entry

Enter the numeric threshold directly in the browser. Confirm the save result. A personal operating threshold is not a certified health limit.

Initial settings

PM2.5: 25 µg/m³; CO2: 1000 ppm; HCHO: 81 ppb. Initial comfort bands: 19–26 °C and 40–60% RH.

Persistence and alerts

Comfort preferences require microSD storage. Alert timing and hysteresis reduce oscillation. No alert does not mean that every substance is under control.

Original Italian display, with English instructions. Photo readings cover partial test periods.

PRINTER 3D



Local configuration

Enter the printer IP, serial number, LAN access code and display name. The LAN access code is not the Bambu account password. One printer is configured at a time.

Available information

Progress, layer count, remaining time, temperatures and filament appear when supplied by the printer. The model name does not automatically select a different protocol driver.

Preview and job name

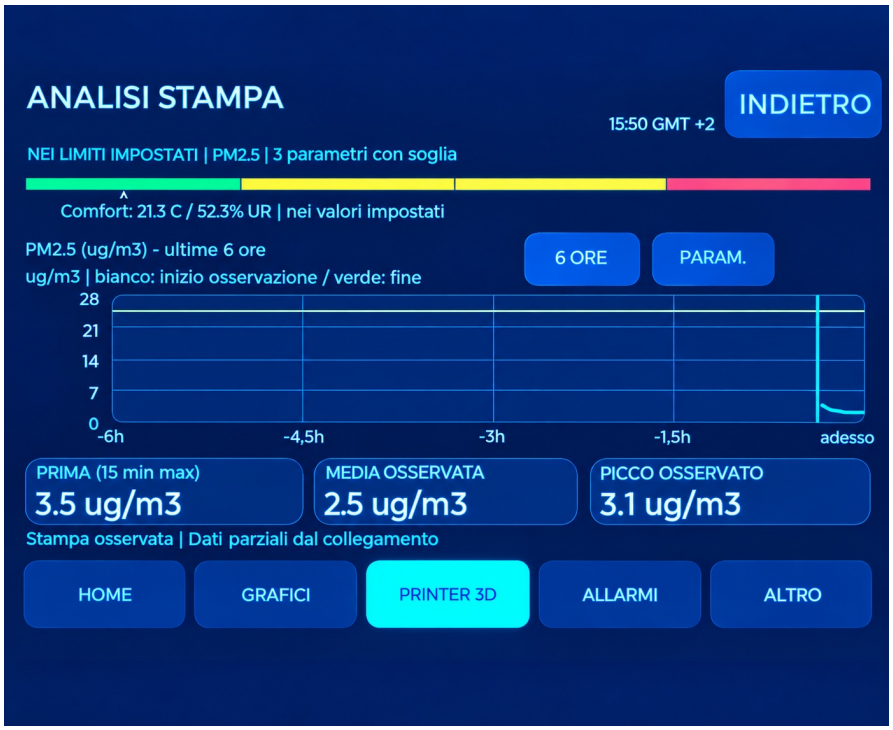
The display preview requires an accessible 3MF. Identical job-name parts separated by “ + ” are displayed once. Different names remain, and the source file is not renamed.

Connection state

Idle or powered-off is distinct from data loss during a print. Old values are identified as stale. Preview retrieval may temporarily suspend and then resume printer data.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Print analysis and environment bar



Session context

Compare the selected environmental parameter with available readings before and during the print, including the session average and peak.

Partial sessions

Starting monitoring after printing has begun, missing samples or restarting makes the comparison partial. Simultaneous variation does not prove the printer caused it.

Summary bar

The bar reflects the worst condition among valid parameters with applicable thresholds. Comfort is separate. It is not a certified health index.

Preserve context

RAM session data is lost on restart. Retain CSV files and printing conditions for later comparisons.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Alert centre



Read the cause

Check the parameter, current value, threshold and data age.

Printer status

A normally idle or powered-off printer is treated differently from loss of communication during printing.

Return to normal

Persistence and a return margin prevent a state change at every small fluctuation.

Informational alerts

GUARDIAN does not stop the printer or replace ventilation and other preventive measures.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Network and browser access



Wi-Fi

Open OTHER → NETWORK. Configure the local network and read the signal strength in dBm, or the disconnected message.

Browser address

Use the address shown by your own GUARDIAN. The credentials and addresses in this guide are examples.

Local access

The local web service uses HTTP. Use a trusted network. For remote access use your network VPN rather than exposing the HTTP port to the Internet.

After changes

Wait for the save result. Connection changes may require a restart; if the IP changes, use the address shown on the display. A guardian-… .local name may not resolve through a VPN.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Browser settings and section order

Display order

Air quality; alert centre; environmental charts; environmental averages; print analysis; 3D printing; SD archive; 3D log download; Wi-Fi connection; Wi-Fi network; printer settings; device access and password; time zone; display lighting; personal comfort; maintenance/calibration; system.

Editing

Thresholds accept direct numeric entry. Available settings include comfort, lighting, time and printer configuration. Check the save result. Device changes synchronise while protecting fields being edited.

Login

Keep your own password private. Remembered access can last up to 30 days depending on session and clock state. Refresh browser cache after a firmware update if an old page remains.

Printer view

Monitoring is read-only for the printer: no movement or stop commands. Preview is on the physical display; GUARDIAN settings remain configurable.

Time zone and display lighting



Time zone

Select a geographic profile and save. Daylight saving follows the selected profile. Synchronisation requires a reachable time source; check the clock after startup.

Manual lighting

Adjust brightness from the display or smartphone browser. Range: 5–100%.

Automatic lighting

Set day/night levels and start/stop times. Defaults: 100% / 15%, night 22:00–07:00. Periods crossing midnight are supported.

Storage

Lighting preferences are on microSD. An invalid clock uses the daytime level. Correct the clock before testing automatic transitions.

Original Italian display, with English instructions. Photo readings cover partial test periods.

microSD reading and test



READ SD

Check capacity, free space, recording status and row count.

TEST SD

Writes, reads, compares and removes a temporary file. Wait for the final result.

Recording

Normally every 30 seconds, with additional events. Interpret errors and dropped records with the serial log.

Removal

Power off before removing the card. Never remove it during writing, testing, downloading or formatting.

Original Italian display, with English instructions. Photo readings cover partial test periods.

CSV download and interpretation

Latest file

Use the dedicated link to the file currently being recorded. Use subsequent archive pages for other CSV files.

Time order

Historical filenames are not necessarily chronological. Use timestamp columns and account for the period before clock synchronisation.

Import

UTF-8 text, semicolon separator and decimal point. Avoid automatic spreadsheet conversions that change numbers or dates.

Fields

Logs include time, uptime, events, validity, environmental values with units, sensor status and alerts. Extended records include print context, partial-session status, job, filament and progress when available.

Retention

An empty field is not zero or proof of no emissions. Back up periodically. Charts do not replace the archive; formatting removes files and SD preferences.

Formatting the microSD

Before formatting

Back up all files and note lighting/comfort settings. FORMAT erases the entire card contents, not just CSV files. Keep power stable.

Request and confirmation

Open the SD page, choose FORMAT and read the confirmation. A request message is not the final result.

Wait

Do not switch off, remove the card or start other SD operations. Wait for formatting and verification to finish.

Verify success

Run READ SD and TEST SD, check that recording resumes and download the new CSV.

If it fails

Retain the display message and GUARDIAN SD FORMAT serial lines. FatFs 3 alone does not prove card failure. Check power and access before retrying.

Effects

SD preference files are erased. Values can remain in RAM until restart; save them again if required. This is separate from GUARDIAN reset and CO2 reset.

Sensor and fan cleaning



Sensor page

Check SEN69C presence and status. Inspect connections only with power off.

Cleaning cycle

Open MAINTENANCE, start the fan cleaning cycle and wait. Measurements pause and then restart.

Meaning of completion

A completed sequence is not an independent test of cleaning effectiveness or fan speed.

Handling

Do not open the sensor or insert objects into its air openings. Follow Sensirion operating and maintenance guidance. [S1, S3]

Original Italian display, with English instructions. Photo readings cover partial test periods.

CO2 preparation and calibration

Known reference

Outdoor CO2 is not a universal fixed value. Do not force calibration using an assumed concentration simply because the unit is outside. Use a reliable reference.

Prepare

Use stable air away from breath and local sources. Follow manufacturer conditions. PREPARE runs for three minutes before READY; that alone does not guarantee equilibrium. [S1]

Confirm

Enter the known reference in ppm within the allowed 400–2000 ppm range. Confirm only after checking the conditions.

Read the result

Wait for completion and measurement restart. If preparation expires or an error occurs, correct the cause and repeat.

Record

Note date, reference, conditions and result. This command does not issue a calibration certificate.

CO2 factory reset



Scope

The command concerns the CO2 channel and the calibration/learning data handled by the manufacturer command.

Procedure

Open CO2 RESET, read the warning, confirm and wait for completion and restarted measurements.

After reset

Allow stabilisation and reassess behaviour. Reset itself is not proof of accuracy.

Separate operations

It does not reset every SEN69C channel, erase CSV files or perform a GUARDIAN factory reset. [S1]

Original Italian display, with English instructions. Photo readings cover partial test periods.

Diagnostics



DISPLAY

Cycle the colours and check uniformity. Distinguish momentary transition artefacts from persistent lines. Document any persistent defect.

TOUCH

Touch different areas and check position/response. CLEAR clears the test view; it does not reset GUARDIAN.

SD

READ, TEST and FORMAT have different effects. Wait for each final result.

Serial log

At 115200 baud inspect uptime, memory, sensor samples, errors and network events. A restarted counter indicates a new boot; printer data loss alone does not prove GUARDIAN rebooted.

Original Italian display, with English instructions. Photo readings cover partial test periods.

Restart, reset and loan preparation

Restart

Restarts GUARDIAN, interrupting RAM history and the active session. Previously written files remain.

Factory reset

After confirmation, clears the settings managed by the firmware, including network, printer, time zone, thresholds and language. Reconfigure and verify browser access, comfort and lighting.

Data on microSD

Factory reset does not automatically erase CSV files or perform CO2 reset. Back up and separately format the card, or use a new card when lending the unit.

Before lending

Use firmware without embedded personal credentials, reset GUARDIAN and verify a fresh configuration is required. Inspect remaining files and job names. Reset/format are logical cleanup, not forensic erasure.

Unresponsive interface

An on-screen reset is unavailable if the interface is frozen. Record the fault and restart physically if necessary.

Troubleshooting

Serial port missing or busy

Try a direct data cable and another USB port. Remove data-line jumpers from the adapter. Close other serial monitors and select the port that appears now.

Display or sensor absent

Check OPI PSRAM, library versions and boot log. With power off, verify sensor 3.3 V, common GND and SDA19/SCL20.

TLS or stale printer data

Check printer power, local IP, network and LAN access code. Bambu Studio via cloud can work even when the local connection fails.

No preview

Retain the log and check the active job and 3MF availability. Missing preview does not mean the print stopped.

CSV or slow listing

Use the current-file link and archive page controls. Check card accessibility and status.

Browser or language outdated

Check the device address, reconnect and refresh cache. Verify the language on SYSTEM. Save edits before changing language.

Measurement quality and documentation

After servicing

Test startup, touch, sensors, SD/CSV, browser, lighting and printer data. Check a new file and increasing row count after SD work.

Keep a configuration record

Archive the IDE, board options, lv_conf.h, library versions and compile summary. Record physical wiring changes.

Validation

Compare with suitable references under documented conditions. Characterise warm-up, enclosure heating and airflow. Software testing does not establish metrological accuracy.

Calibration document

The attached Sensirion statement concerns factory calibration of the listed sensor families. It is not an individual calibration report for the complete GUARDIAN device. [S2]

Product conformity

Component REACH/RoHS documentation does not certify the assembled product. RED, RoHS, WEEE and REACH are references for an eventual finished-product assessment; no completed GUARDIAN certification is claimed. [S11–S16]

Display lighting: validated update

25 kHz backlight PWM

The owner reports that the 5 kHz whine disappeared after increasing the backlight PWM to 25 kHz. Manual brightness and the automatic day/night schedule were tested successfully.

Version identification

This edition describes bilingual GUARDIAN with the 25 kHz modification. The tested sketch is labelled 1.0.0-rc3-pwm25-test1; this label may still appear on the device and in the browser.

Settings and updates

Brightness remains adjustable from the display and browser. This change requires neither factory reset nor SD formatting.

Scope of the test

Functional testing on the owner’s prototype supports keeping this setting. It is not a metrological validation or a certification of the complete product.

The OTHER menu



Select SYSTEM (SISTEMA) to choose the interface language or access restart and reset.

Retouched from the supplied photograph: reflections removed and perspective corrected. Original Italian labels retained.

System and interface language



Select Italiano or English. The choice is saved without restarting. Restart preserves settings; factory reset clears GUARDIAN settings and web access. CSV files remain on microSD.

Retouched from the supplied photograph: reflections removed and perspective corrected. Original Italian labels retained.

Individually tested by Sensirion

Factory calibration and individual testing

Sensirion states that each listed device is individually tested after calibration. SEN69C is included in the SEN5x/SEN6x declaration. [S2]

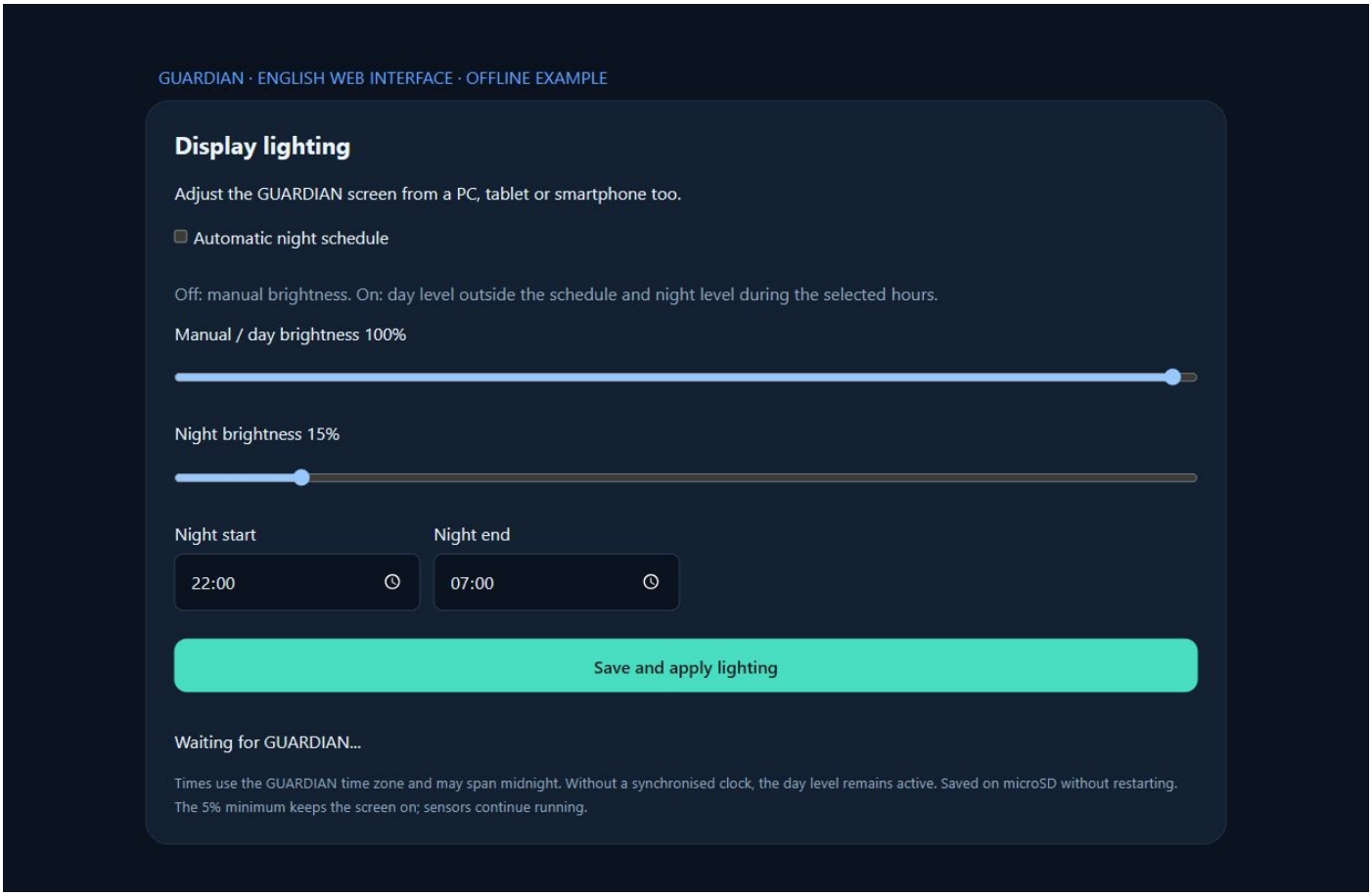
Scope of the statement

The May 2026 declaration is titled Particulate Matter Sensors. It must not be presented as certification of every gas channel or of the complete GUARDIAN device.

Manufacturer evidence

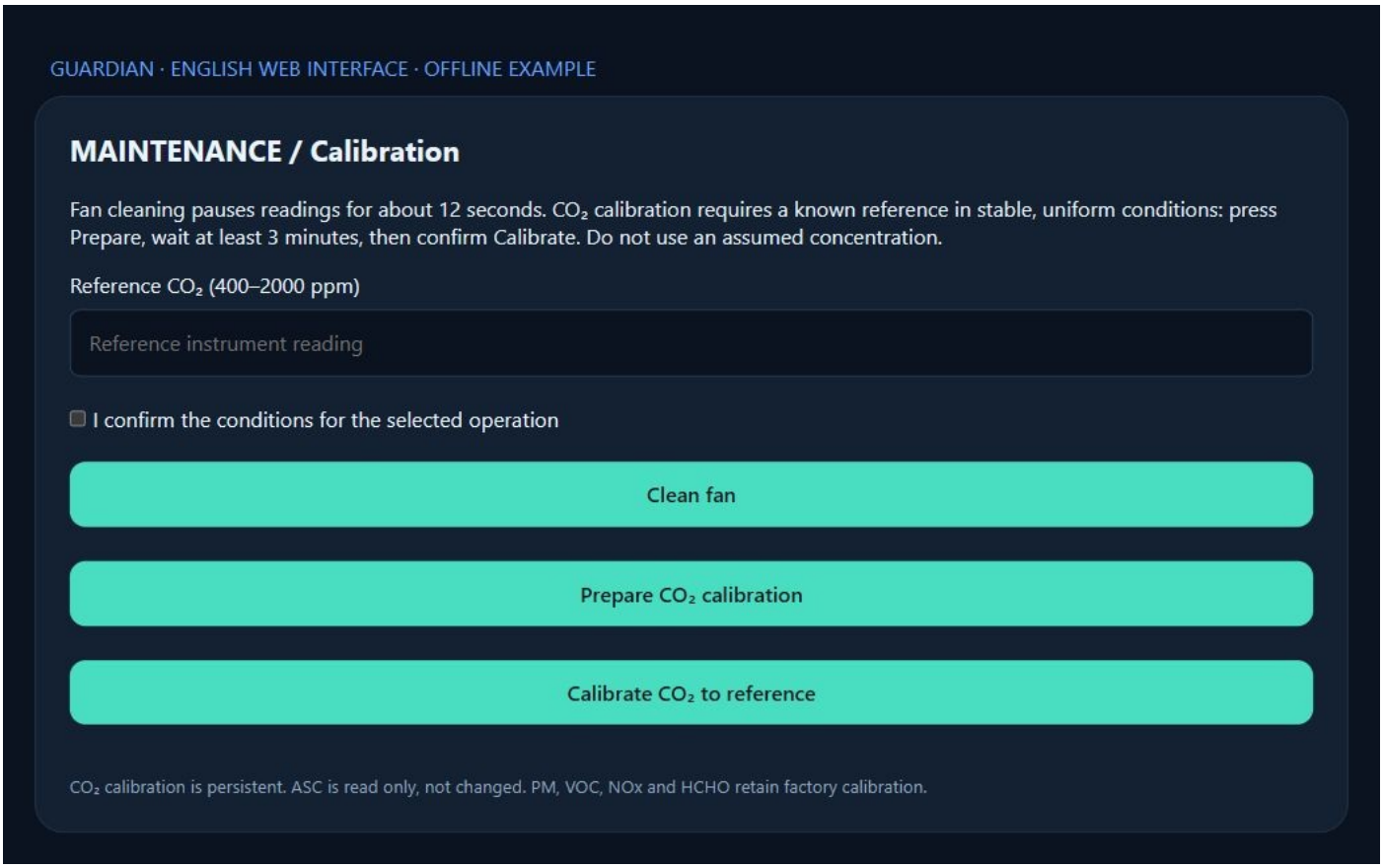
The original statement is attached to the manual. Enclosure integration, airflow and operating conditions still require their own checks.

English interface: display lighting



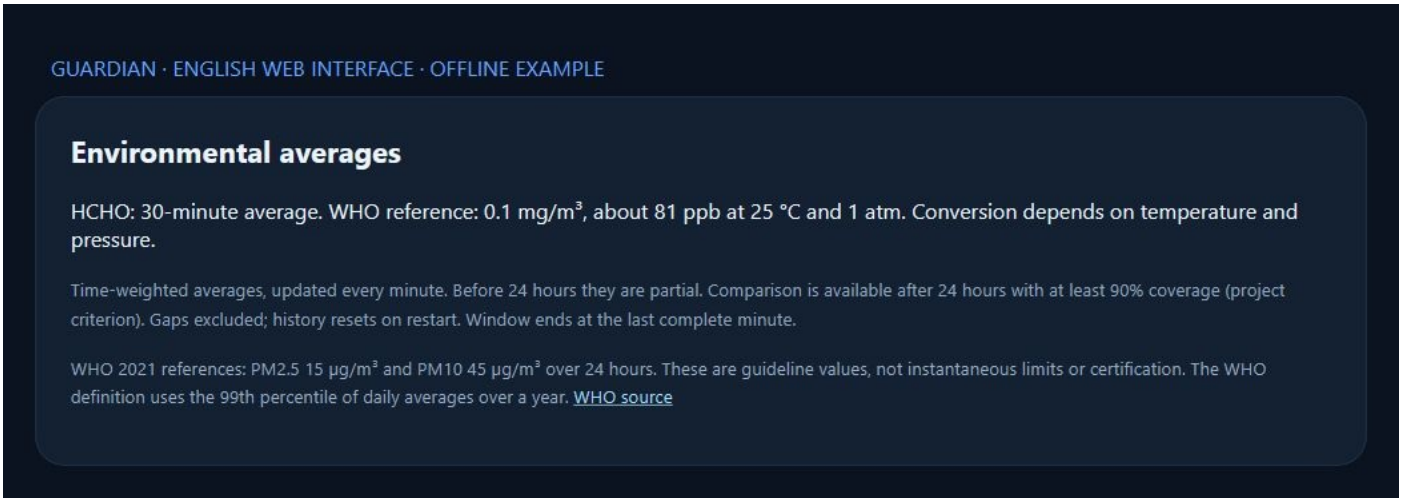
Offline example rendered from the English firmware web page. No live readings or personal credentials. The layout adapts to browser width.

English interface: maintenance



Offline example rendered from the English firmware web page. No live readings or personal credentials. The layout adapts to browser width.

English interface: averages



Offline example rendered from the English firmware web page. No live readings or personal credentials. The layout adapts to browser width.

Sources and reference documents

S1 - Sensirion SEN6x datasheet
Open original source

S2 - Sensirion factory calibration statement, SEN5x/SEN6x
Open original source

S3 - Sensirion SEN6x mechanical design and assembly
Open original source

S4 - Sensirion sensor specification statement
Open original source

S5 - Sensirion temperature compensation
Open original source

S6 - WHO Global Air Quality Guidelines, 2021
Open original source

S7 - WHO indoor air quality: selected pollutants, 2010
Open original source

S8 - EPA research on 3D-printing emissions
Open original source

Manufacturer and regulatory references do not certify the complete GUARDIAN device. Original documents retain their source language.

Sources and reference documents

S9 - EPA Air Sensor Toolbox
Open original source

S10 - UBA: CO2 and ventilation
Open original source

S11 - Radio Equipment Directive 2014/53/EU
Open original source

S12 - RoHS Directive 2011/65/EU
Open original source

S13 - Delegated Directive (EU) 2015/863
Open original source

S14 - WEEE Directive 2012/19/EU
Open original source

S15 - REACH Regulation (EC) 1907/2006
Open original source

S16 - Sensirion REACH/RoHS declaration
Open original source

Manufacturer and regulatory references do not certify the complete GUARDIAN device. Original documents retain their source language.

Sources and reference documents

S17 - Espressif Arduino ESP32 installation
Open original source

S18 - Freenove FNK0115 repository
Open original source

S19 - Sensirion SEN69C product information
Open original source

Manufacturer and regulatory references do not certify the complete GUARDIAN device. Original documents retain their source language.

SENSIRION
THE SENSOR COMPANY

Calibration Certification

Name and address of the manufacturer: Sensirion AG
Laubisruetistrasse 50
CH-8712 Switzerland

Description: Particulate Matter Sensors

- SEN50, SEN54, SEN55
- SEN62, SEN63C, SEN65, SEN66, SEN68, SEN69C

The above mentioned products are calibrated to meet the specifications according to the corresponding Sensirion data sheet. Each device is individually tested after its calibration.

Sensirion uses transfer standards for the calibration. These transfer standards are themselves subject to a scheduled calibration procedure. The calibration of the reference itself used for the calibration of the transfer standards is performed by an ISO-9001:2015 registered calibration system.

The accreditation body is full member of the International Laboratory Accreditation Cooperation (www.ilac.org). Calibration certificates issued by facilities accredited by a signatory to the ILAC Mutual Recognition Arrangement (MRA) are accepted by all signatories to the ILAC MRA.

This provides traceability of measurement to recognized national standards and to units of measurement realized at the "National Physical Laboratory" (NPL) or other recognized national standards laboratories like "Physikalisch-Technische Bundesanstalt" (PTB) or "National Institute of Standards and Technology" (NIST).

Staefa, May 2026

Stephan Weber,
Director, Head of Quality Management, Sensirion AG

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